

AI Summer School 2026: Team Project

June 12, 2026

University of Pittsburgh

Introduction and Problem Statement:

Knee osteoarthritis is one of the most common musculoskeletal (MSK) disorders and a leading cause of pain and disability worldwide. One of the primary radiographic indicators of osteoarthritis progression is narrowing of the knee joint space, which reflects the loss of cartilage between the femur and tibia. Accurate measurement of the joint space can help clinicians assess disease severity and monitor disease progression over time. However, manually identifying and outlining the joint space can be time-consuming and subject to variability between observers. Automated segmentation of the knee joint space provides a foundation for many downstream AI and clinical applications, including joint space width measurement, osteoarthritis severity assessment, disease progression monitoring, surgical planning, and quantitative MSK imaging analysis.

In this team project, you will develop an AI-based segmentation system to automatically identify the knee joint space on knee radiographs. You will train and evaluate a U-Net model using a dataset of 500 knee X-ray images with corresponding segmentation masks.

The goal is not just to build a model - but to experience the entire medical AI pipeline: dataset exploration, preprocessing, model training, evaluation, visualization, and communication of the results.

Good luck and have fun!

Team Creation:

In-Class: You will form groups of 3-4 students.

Remote: You will form groups of 3-4 students.

Project Tasks

Task 1: Dataset Exploration

- Download the dataset from the GitHub repository

- Visualize several image-mask pairs
- Verify image resolution
- Determine whether this is a binary or multiclass segmentation problem
- Discuss any challenges that may affect segmentation performance

Deliverable: Include 2-3 example image-mask pairs and discuss any challenges you observe in the dataset in your presentation.

Task 2: Dataset Preparation

- Split the dataset into training, validation, and test sets using 70% for training, 15% for validation, and 15% for testing
- Setup the appropriate preprocessing needed for the dataset
- Setup the PyTorch Dataset to load the images and masks.

Deliverables: Describe your preprocessing pipeline and report the number of images in each dataset split.

Task 3: Model Training

- Make sure the GPU for Google Colab is enabled
- Implement and train a U-Net segmentation model
- Experiment with different:
 - Learning rates
 - Numbers of training epochs.
 - Encoder backbones, using at least 2 different backbones
- Choose the best performing model for each backbone from your experiments

Deliverables: In your presentation

- Report the learning rate, number of epochs, loss function, and encoder backbone for each experiment
- Provide training and validation loss curves
- Identify the best-performing model for each backbone

Task 4: Model Evaluation

- Evaluate the best-performing model (based on your validation results) for each encoder backbone on the test set
- Report mean Dice and Intersection over Union (IoU) and class-specific Dice and IoU

Deliverable: In your presentation report Dice and IoU metrics, briefly discuss model performance, and provide at least 2 examples where your model was successful and 1 where it failed.

Task 5: Result Analysis

- Discuss:
 - Where does the model perform well?
 - Where does the model struggle?
 - What image characteristics appear to affect performance?
 - How could performance be improved?
 - How did the choice of encoder backbone affect performance and training time?

Deliverable: A short discussion slide summarizing findings and limitations.

Task 6: Thinking about Downstream Tasks (Future Work)

Segmentation is often an intermediate step in a larger AI pipeline.

How could the segmentation masks generated by your model be used to automatically measure joint space width or assist in osteoarthritis severity assessment?

Deliverable: Be prepared to discuss your ideas during your presentation.

Team Presentation

- Prepare a short team presentation that includes:
 - Background & Clinical Motivation (1 slide)
 - Dataset Exploration (1 slide)
 - Dataset Preparation (1 slide)
 - Methods (1-2 slides)
 - Results (1-3 slides)
 - Conclusions & Future Work (1 slide)
- Presentation Time: 1:00PM – 3:15PM
- Each team will have approximately 5 minutes.

Assistance:

If you have questions at any time, please reach out to:

- Nick Littlefield
- Fengyi Gao

GitHub Repository:

All datasets and the example code from the U-Net session is available at:

<https://github.com/pitthexai/AISummerSchoolinMedicalImagingInformatics>